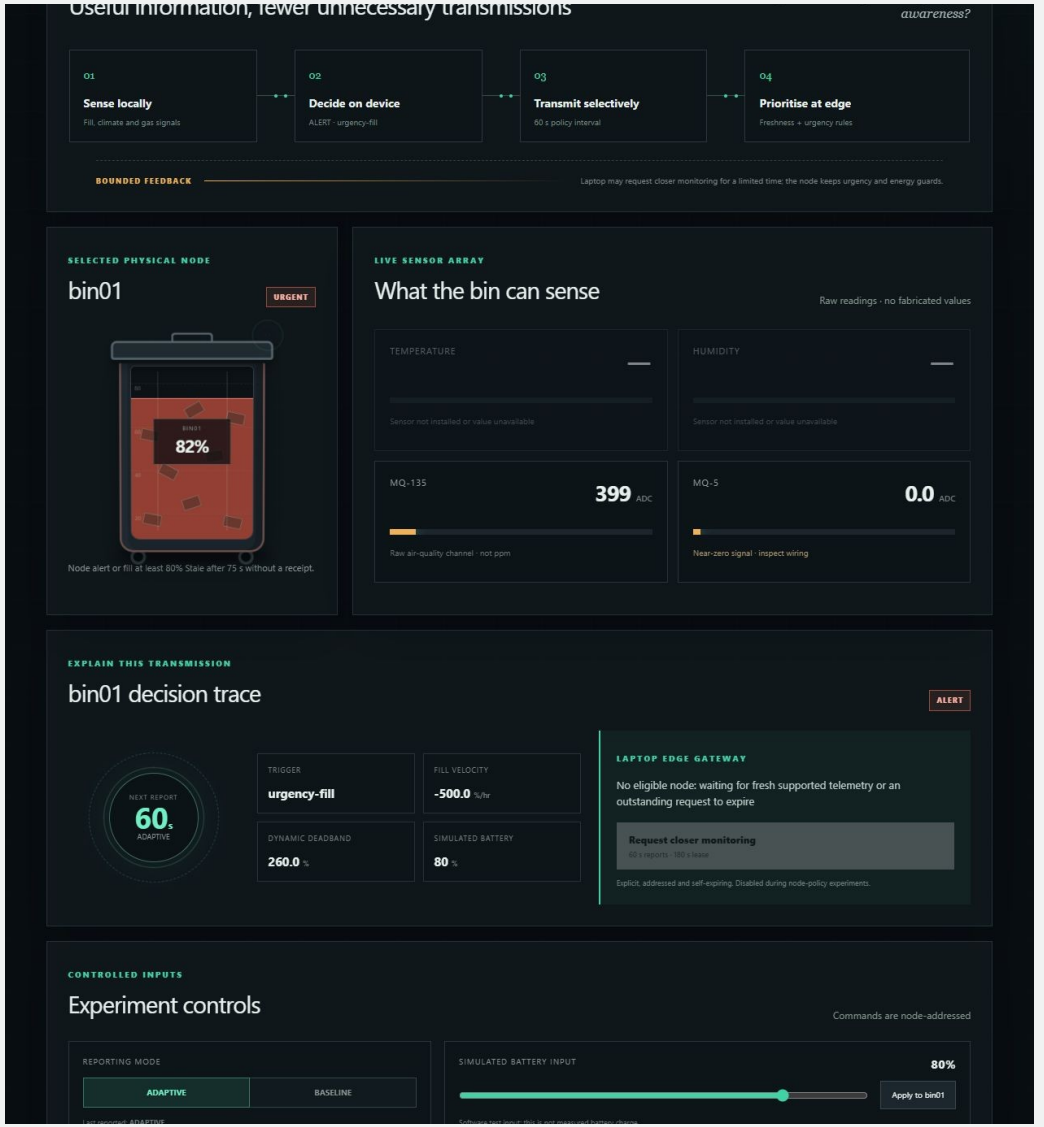
Edge-Based Energy-Efficient IoT Smart Waste Management

A functional two-node edge prototype with selective reporting, laptop gateway control and an evaluation-ready evidence pipeline.

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RESEARCH PROBLEM

Periodic reports spend energy even when nothing changes

01

Fixed reporting

Conventional nodes transmit on a fixed timer regardless of whether the bin state has changed.

02

Repeated information

Stable readings consume radio time, gateway capacity and battery energy without changing the collection decision.

03

Operational trade-off

Reporting less saves communication effort, but urgent fill and fault events must still arrive on time.

Research objective

Reduce unnecessary transmissions through local edge decisions while preserving timely alerts and useful operational data.

SYSTEM ARCHITECTURE

The prototype closes the loop across five functional units

01

Physical nodes

ESP32 bin01
ESP-12E bin02
Fill + climate

→

02

On-device policy

FSM · AoI timer
Deadband · AIMD
Urgency rules

→

03

MQTT transport

Node topics
Validated payloads
Local broker

→

04

Laptop edge

Freshness + priority
SQLite evidence
Bounded feedback

→

05

Dashboard

Live bins
Decision trace
Experiment controls

←

BOUNDED FEEDBACK

The gateway may request closer monitoring for a limited time; the node keeps its safety and energy guards.

Information value triggers each transmission

Four operating states

ALERT	60 s	≥80% fill, gas rule or rapid filling
ACTIVE	300 s	≥60% fill
STABLE	300–1800 s	AIMD adapts the interval
CONSERVE	900 s	≤30% battery input

Three transmission gates

1 Urgent state entry

Immediate evidence when the node first enters an urgent condition.

2 Age-of-information timer

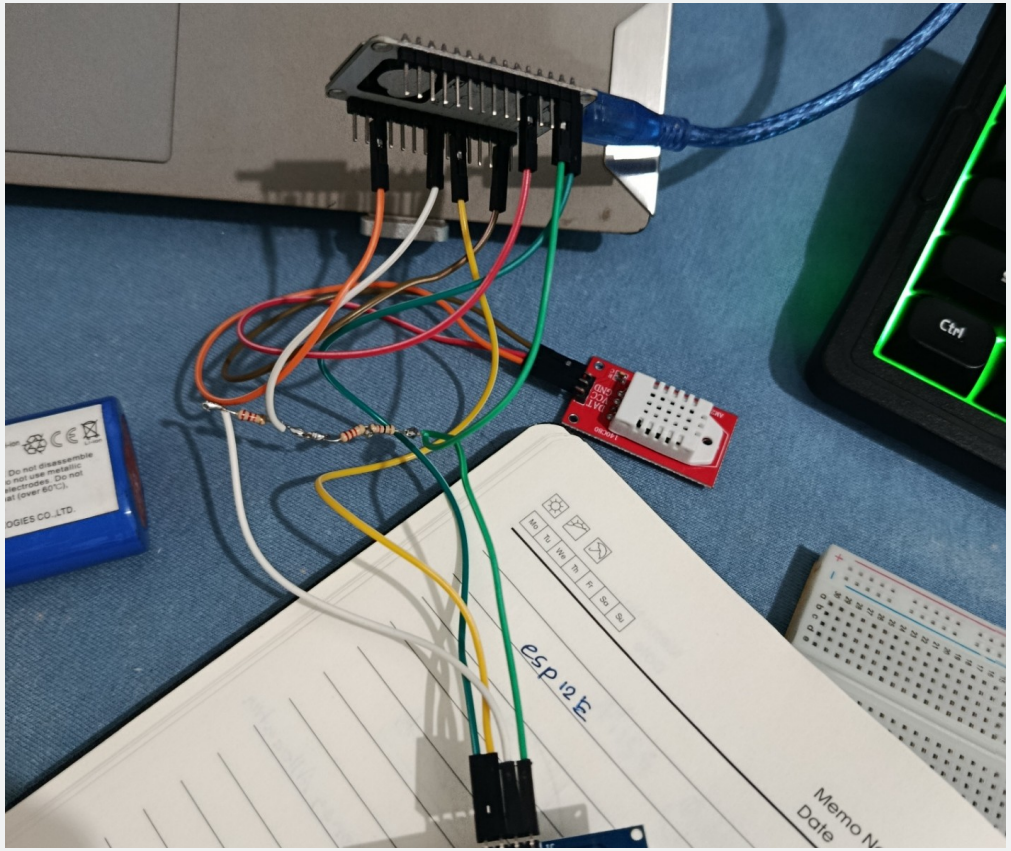
A maximum interval prevents the gateway view from becoming stale.

3 Dynamic fill deadband

Meaningful fill change can trigger a report before the timer expires.

Transmit = urgent entry OR timer OR deadband crossing

Two physical nodes report with explicit sensor status



BIN01 · ESP32

HC-SR04 fill · DHT22 climate · MQ-135 raw air-quality channel

Corrective action: remove MQ-5 from the decision path; verify or replace the intermittent DHT22.

BIN02 · ESP-12E

HC-SR04 fill · DHT22 temperature and humidity

No gas sensor by design. Echo is protected by the available resistor divider.

Research integrity

Unavailable or uncalibrated values remain visible as limitations; the system does not invent replacements.

The laptop performs real edge-gateway work

01

Validate and preserve

Checks topic, node identity, schema and payload limits before storing accepted telemetry in SQLite.

02

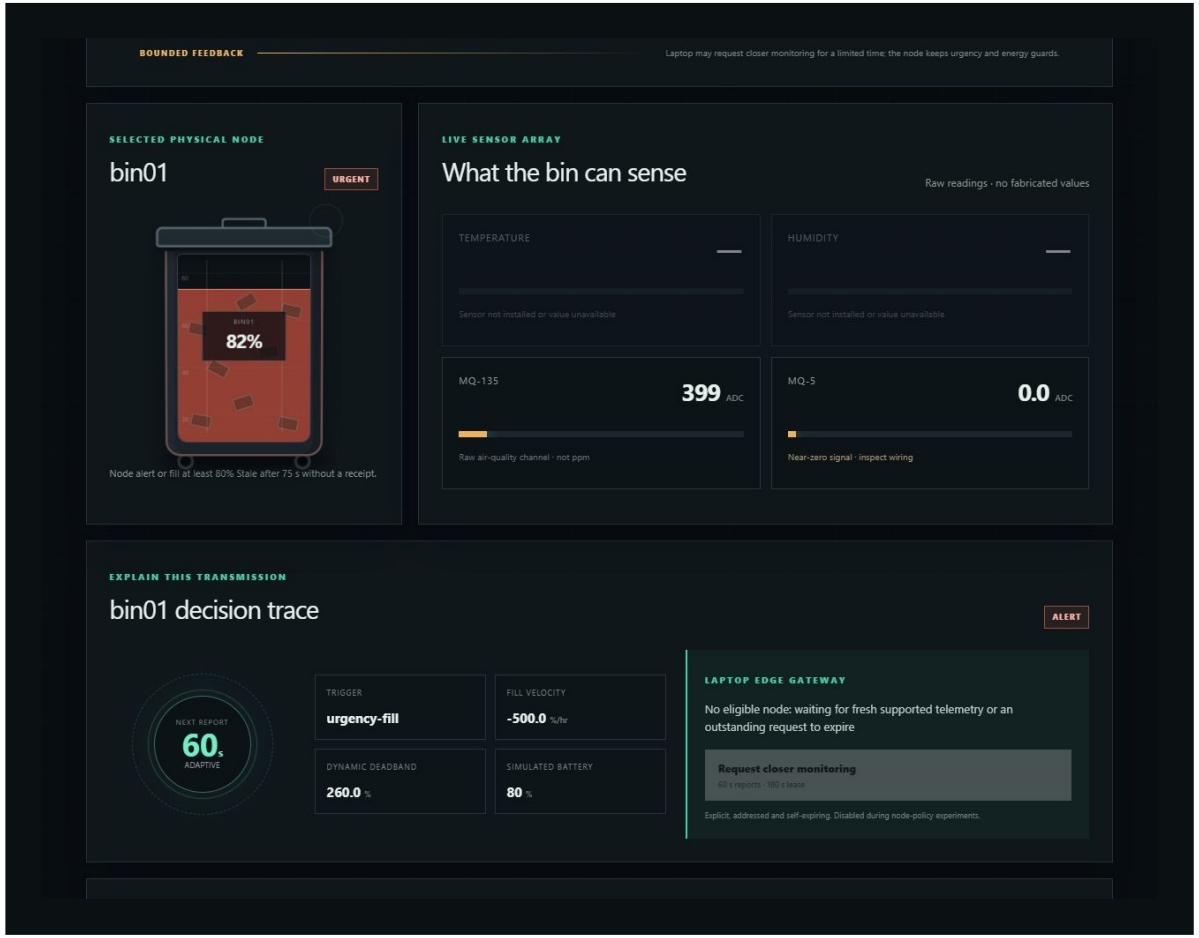
Prioritize across nodes

Combines freshness, state and sensor health to select the node that needs attention.

03

Request bounded feedback

Sends a node-addressed monitoring request with a lease and records the acknowledgement.

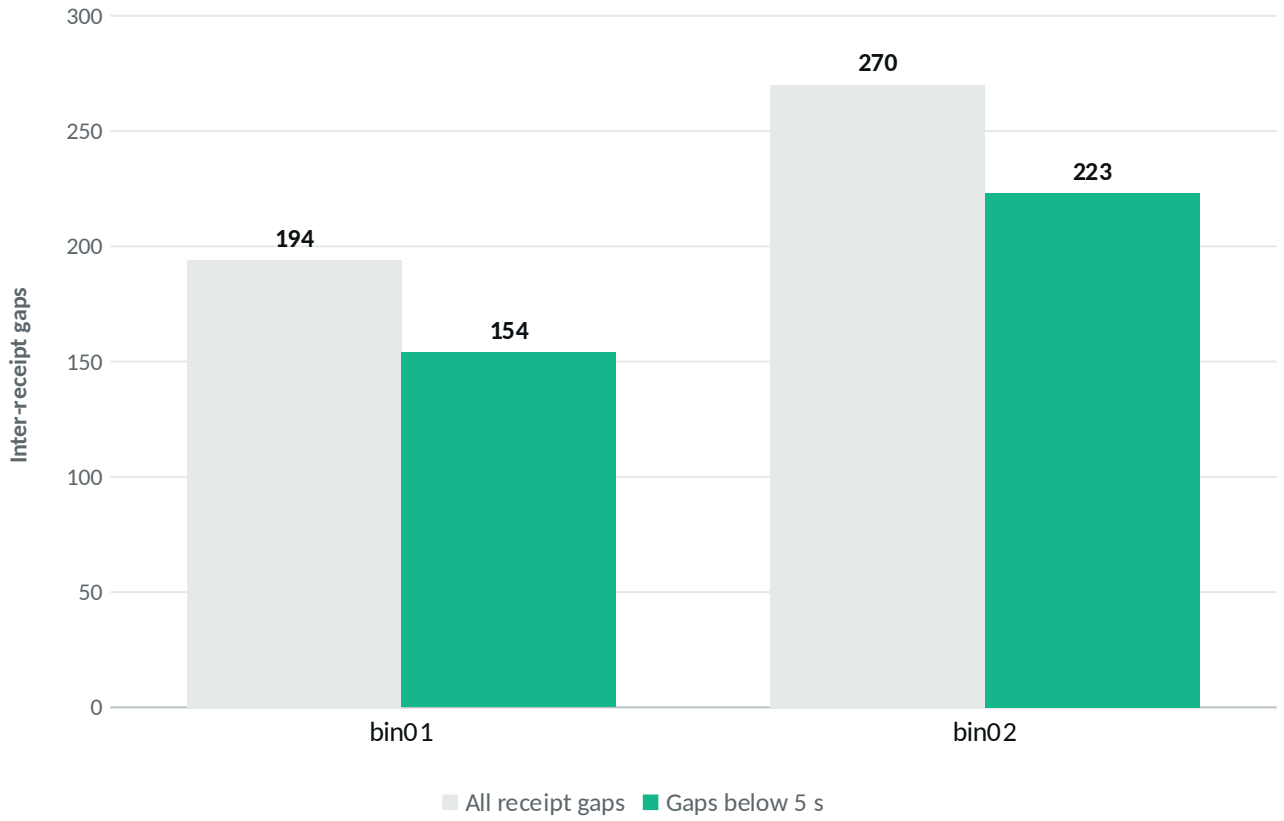


Implementation is functional; evaluation is the remaining gap

FUNCTIONAL UNIT	STATUS	EVIDENCE / NEXT ACTION
Two physical sensing nodes	FUNCTIONAL	Live fill/climate telemetry from bin01 and bin02
On-device edge policy	FUNCTIONAL	FSM, AoI timer, deadband, AIMD and urgency rules
Laptop gateway + evidence store	FUNCTIONAL	Validation, SQLite, priority, bounded command/ACK
Operator dashboard	FUNCTIONAL	Animated bins, sensor status, policy trace and run labels
Energy evaluation	NEXT	Install INA219 on both nodes; run matched baseline/adaptive trials
Mobile, cloud, routing, weight	PARTIAL	Implement later or agree a formal scope revision with supervisor

Completion is reported by evidence, not an invented percentage.

Diagnostics proved integration—and exposed burst behaviour



79%

bin01 gaps below 5 seconds

83%

bin02 gaps below 5 seconds

Interpretation

Movement and noisy fill changes repeatedly crossed the event path. The nominal state interval alone does not cap traffic.

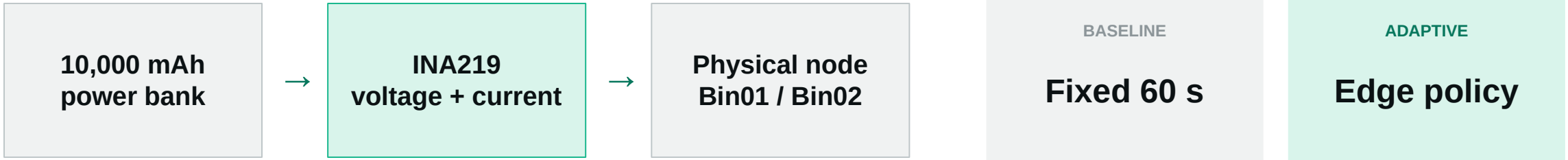
Research decision

Treat this as diagnostic evidence—not as a transmission-reduction result. Add cooldown, hysteresis and matched trials.

Window: approximately 25 minutes per node · active diagnostic run

CONTROLLED EVALUATION

INA219 measurements will turn the claim into evidence



Same node · same sensor scenario · equal duration · repeated trials

Primary measures

TX
Transmission-count reduction

mWh
Measured node-input energy

ms
Alert latency and recall

AoI
Freshness and fill error

Hypothesis: adaptive reporting reduces transmissions and communication energy without unacceptable loss of alert timeliness or data freshness.

The system is designed around operational decisions

01

See what needs attention

Operators view fill condition, freshness, climate and supported air-quality signals in one place.

02

Respond to urgent bins

Local urgency rules preserve fast reporting when fill or sensor conditions require intervention.

03

Retrofit existing assets

Low-cost Wi-Fi nodes and a portable edge service can be attached to bins without replacing the complete collection system.

Deployment path

Prototype → **Controlled pilot** → **Secure local deployment** → **Fleet integration**

Benefits are expected outcomes until energy, service and collection trials measure them.

EXECUTION ROADMAP

A staged plan closes the remaining proposal gaps

01 SEP

PP2

Demonstrate two nodes, edge logic, gateway and dashboard

Working prototype

WEEK 1

Stabilize

Remove MQ-5 path, verify DHT22, add health and trigger provenance

Per-sensor health

WEEKS 2–3

Measure

Install INA219, run repeated baseline/adaptive and fault scenarios

Measured evidence

BY 23–28 OCT

Close scope

Paper/thesis evidence; implement or formally re-scope mobile, route and cloud

Supervisor-approved scope

Priority decision

Complete the energy experiment before adding prediction or route optimisation; stronger evidence is worth more than another unvalidated feature.

The edge prototype works. The next result must be measured.

FUNCTIONAL

2 physical nodes

Local decisions + live telemetry

EXPLAINABLE

Visible decision path

State · interval · trigger · feedback

EVALUATION-READY

INA219 experiment

Baseline versus adaptive evidence

Live demonstration: bin state → node decision → MQTT → gateway → dashboard → bounded feedback

PANEL EVIDENCE MAP

Appendix · Evidence mapped to the PP2 assessment

AREA	WEIGHT	DECK / DEMO EVIDENCE
LO1 · problem, gap, units	**10%**	Slides 2–3: research tension and five functional units
LO2 · technical knowledge	**20%**	Slides 4, 8–9: policy, critical diagnosis and experiment design
LO3 · implementation	**40%**	Slides 3–7 + live demo: nodes, gateway, database and dashboard
LO4 · communication / Q&A	**15%**	One traceable story, backup evidence and appendix answers
LO5 · user / commercial value	**15%**	Slide 10: operator outcomes and controlled deployment path

The 40% implementation section is supported by the live system; research claims remain bounded by measured evidence.

Appendix · Policy values and evidence boundaries

Current configuration

Fill ALERT	**≥80%**
Fill ACTIVE	**≥60%**
Rapid fill	**>10% per hour**
Baseline	**60 seconds**
Stable adaptive	**300–1800 seconds**
Deep sleep	**Disabled for live demo**

Evidence boundaries

Battery	Software-controlled percentage; real energy measurement pending
Gas	MQ-135 raw ADC only; not calibrated ppm or a safety alarm
MQ-5	Unstable channel; removal from decision path
DHT22	Bin01 intermittent; health check and replacement planned
Security	Trusted local network; no deployment-grade authentication/TLS
Scale	Two physical nodes; synthetic nodes must be labelled separately

Appendix · Direct answers to likely panel questions

01	**Why is there no Raspberry Pi?**	The laptop currently implements the same logical edge-gateway role. Hardware migration does not change the node/gateway protocol.
02	**Where is the energy-saving result?**	It is not claimed yet. INA219-backed matched trials are the next controlled evaluation.
03	**Why remove MQ-5?**	The signal was unstable. Removing it prevents false alerts; MQ-135 remains a raw, uncalibrated prototype channel.
04	**What is the research contribution?**	An explainable edge reporting policy evaluated against transmission, energy, alert and freshness metrics.
05	**What remains from the proposal?**	Energy measurement first; mobile, routing, cloud, weight and prediction will be implemented or formally re-scoped with the supervisor.